

Indian Institute of Technology Kharagpur

Basic Thermodynamics (ME21202)

(Mid, Spring Semester 2023-24)

Time: 2 hrs

Date: 21/02/2024

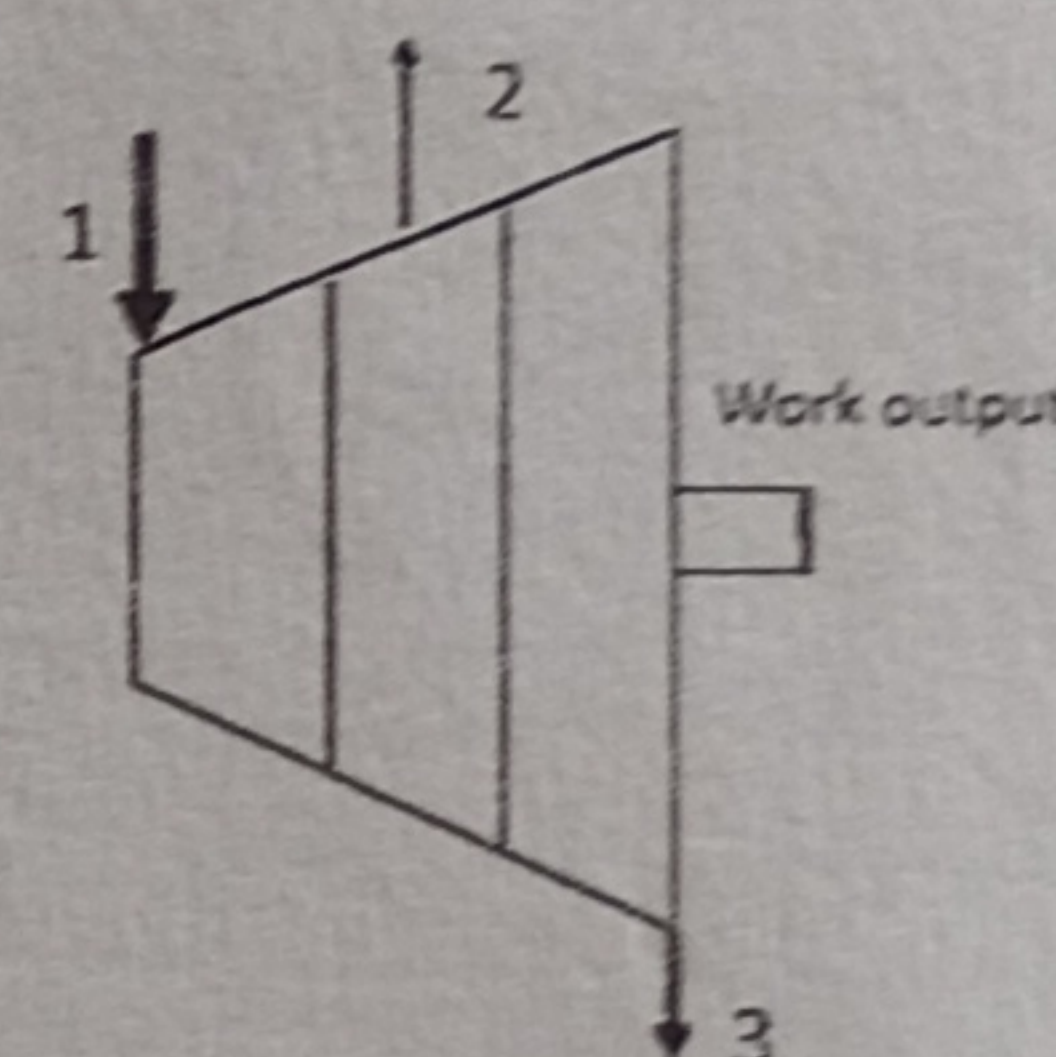
Total marks: 60

Instructions:

1. This question paper contains 5 questions, 3 printed pages (2 pages of property tables of water)
2. Attempt all questions. Attempt each question on a new page.
3. All calculations must be shown in the solution of the problem.
4. Clearly put the final answers inside a box.
5. State any assumptions used clearly.
6. Draw plots wherever needed. You must clearly show liquid-vapour dome, wherever applicable.
7. In case you feel that any data is missing, assume the data and proceed with the solution.

Assume air as Ideal gas: R (specific gas constant) = 0.287 kJ/kg-K, C_p = 1.004 kJ/kg-K, C_v = 0.717 kJ/kg-K

1. A steam turbine receives **steam** at 1.5 MPa, 600°C (state 1) at a rate of 100 kg/s, as shown in the figure. At the middle of the turbine, 30 kg/s is withdrawn at 0.8 MPa, 350°C (state 2). The rest of the steam exits the turbine at 150 kPa with 95 % quality (state 3). Assuming adiabatic turbine, find the total work done by the turbine (kW). (10)



2. A piston cylinder contains **air** at 600 kPa, 17°C and initial volume of 0.01 m³. It undergoes a constant pressure process and does +54 kJ work.
 - i. Find the final temperature of air (in °C). (5)
 - ii. Find the heat transfer to the cylinder (in kJ). (5)

3. a. A commonly used pressure cooker has a lid which is tightly attached to the chamber. The lid has a small opening of $A = 5 \text{ mm}^2$. This opening is covered with a petcock which can be lifted to allow steam to escape. What should be the mass (in kg) of the petcock in order for the water inside to boil at 130°C? Assume atmospheric pressure, $P_o = 100 \text{ kPa}$ and $g = 9.81 \text{ m/s}^2$. (Assume only water present inside the pressure cooker.) (5)

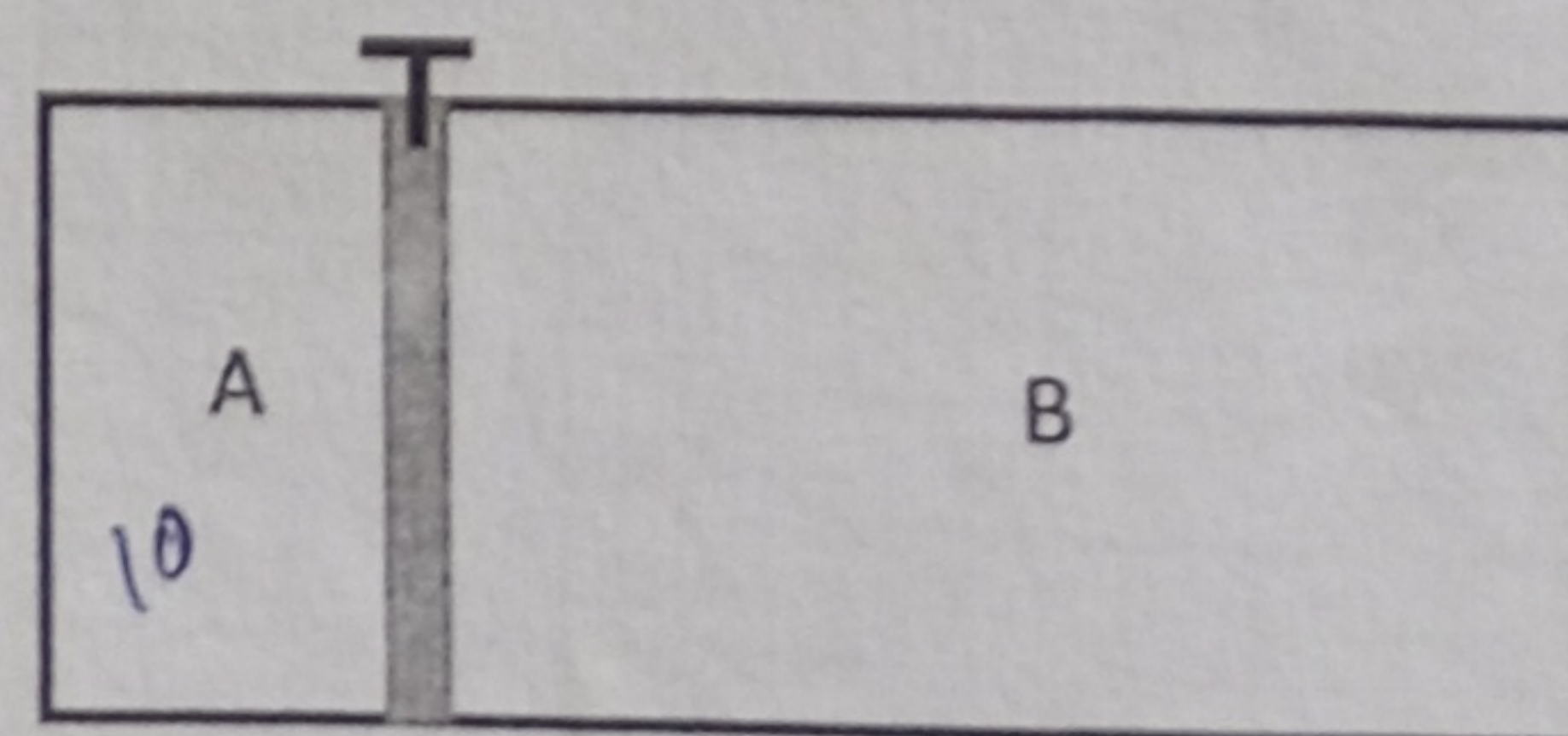
b. Saturated water vapour (100% quality) at 200 kPa is contained inside a piston-cylinder arrangement. At equilibrium, the piston is 0.1 m from the bottom of the cylinder. The cylinder is now heated and the temperature is raised to 200°C.

- i. What is the final distance of the piston from the bottom of the cylinder? (3)
- ii. Draw the process diagram on the P-v diagram. Label the points clearly. (3)
- iii. Draw the process diagram on the T-v diagram. Label the points clearly. (3)

4. A piston-cylinder contains **water** at 500°C, 3 MPa. This is now cooled by polytropic process ($PV^n = \text{constant}$) to 200°C, 1 MPa.

- i. Determine the polytropic exponent (n) for this process? (4)
- ii. What is the specific work done (kJ/kg) during the process? (4)

5. A closed cylinder is divided into 2 chambers by a frictionless piston, which is initially held at its place through a pin. Chamber A occupies a volume of 10 L and contains **air** at 100 kPa, 30°C. Chamber B occupies a volume of 300 L and has **saturated water vapour** at 30°C. Once the pin is removed, both the chambers come to equilibrium at 30°C wherein water goes into a 2 phase state. Consider chambers A and B as the entire system and answer the following:



- i. What is the work done (kJ) by the entire system? (3)
- ii. What is the heat transfer (kJ) to the system? (15)